

Scalable Computer System Design

ECE 750 T39

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Course Overview

What this course is about?

This course focuses on the “scalability” aspects of large-scale software design in practice; it is organized to help the audience,

- Comprehend scalability demands of large-scale problems
- Learn state-of-the-art scalable algorithmic techniques and systems approaches in key problem classes
- Familiarize with relevant computing platforms and tools, and their applications to scalable solution design

Recommended background preparation

It is expected that students have already completed undergraduate computer science/engineering courses that covered the following topics: data structures, algorithms, object-oriented programming, operating systems, computer architecture, computer networks and parallel/distributed computing. Although not required, having taken one of the following courses, ECE 454, ECE 459 or ECE 751, would certainly help, especially, with the programming assignment and term project.

Learning resources

Lecture slide decks, research articles for in-class discussions and writing reviews, and other reading materials will be provided by the instructor. While no formal text book is required, students may find the following book useful:

- Principles of Computer System Design: An Introduction, Jerome H. Saltzer, M. Frans Kaashoek

This course will use LEARN, as well as Piazza as a discussion forum. Important announcements will be posted on LEARN, including a signup link for Piazza. Please use Piazza exclusively to communicate with the teaching team regarding course materials.

Evaluation structure

- Assignments 15%
 - A short programming assignment 7% (group)
 - Writing paper reviews 8% (individual, two reviews, one page each)
- Two in-class tests 30% (individual, closed book, 45 minutes each)
- Leading paper discussion and class participation 10%
- Course project 45% (group)
 - Project proposal 5%
 - Project presentation 20%
 - Project report 20%
- 65% is the minimum passing grade for Master's students
- 75% is the minimum passing grade for PhD students

Course project

- A term-long group project that exercises the steps involved in designing, implementing and evaluating a scalable computing solution (software, hardware or co-design).
- Student groups have the flexibility of choosing the problem, hardware platform (parallel shared memory or shared-nothing distributed) and implementation framework.
- There are three deliverables:
 1. Project proposal - a brief description of the problem of interest, its scalability requirements, overview of anticipated solution design, research methodology and work plan.
 2. Project presentation - a 20-minute group presentation of the completed project.
 3. Project report - problem definition and context, scope of contribution, solution design, evaluation methodology and results, and future work.

Depending on the class size, each group will have 3 to 5 members.

ECE Linux computing facility

Programming assignments and projects should be completed using the Linux environment. The ECE department provides several Linux servers for this purpose:

- eceterm1, eceterm2, eceterm3 are used only for accessing the other machines from outside the campus firewall
- eceubuntu1, eceubuntu2, eceubuntu4
- ecetesla0, ecetesla1, ecetesla2, ecetesla3, ecetesla4
- ecehadoop cluster, ece-nebula GPU cluster

Note 1: See <https://ece.uwaterloo.ca/Nexus/arbeau/clients/> for load statistics.

Note 2: Use TCP port numbers in the range 10,000-11,000.

Note 3: To speed up access to ecelinux hosts, please use SSH public key authentication, set up an SSH agent and enable agent forwarding.

Contact information

Instructor:

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Teaching Assistants:

TBD

Note: Please include “ECE 750 T39” in the subject line of emails sent to the teaching team.